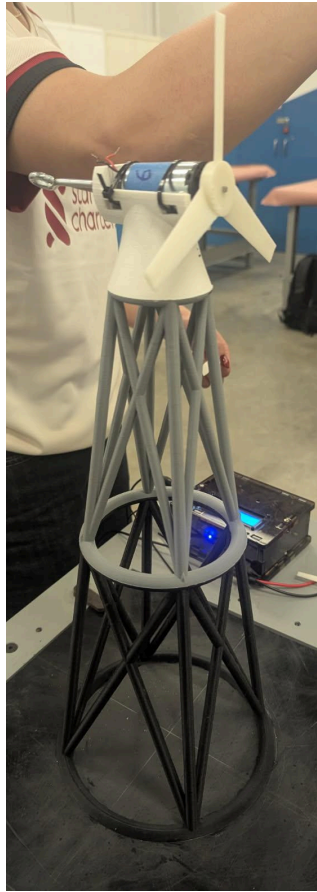


Wind Turbine Final Report



University of California, Mechanical Engineering Dept. E26

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Project Summary

This project focused on the design, fabrication, and testing of a small-scale wind turbine system for the University of California, Mechanical Engineering Department E26 course. The main objective was to create a wind turbine tower with high stiffness under applied loading and rotor blades capable of generating maximum power under a 25 mph wind condition. The project required our team to apply the engineering design process through initial sketches, SolidWorks CAD modeling, 3D printing, finite element analysis, assembly, and experimental testing.

The wind turbine consisted of two main custom-designed components: the tower structure and the rotor blades. The tower was designed to support the generator at the required height of 16 inches from the platform to the center of the motor shaft while remaining within the 17 in³ material volume constraint. Because the available 3D printers had an 8 x 8 x 8 inch print volume, the tower body was divided into three printed sections and assembled using strong adhesive. The final tower design combined ideas from multiple team sketches, with the goal of improving stiffness while maintaining manufacturability and satisfying all design constraints.

The rotor blades were designed to maximize power generation while staying within the required 6-inch swept diameter. The team considered blade shape, airfoil profile, thickness, angle of attack, and twist angle when developing the final blade geometry. During testing, the blower was adjusted to produce approximately 25 mph wind, and the electrical load was varied to find the operating condition that produced the maximum output. The final turbine generated a maximum power output of **1.003 W**, corresponding to an efficiency of **1.6%**.

The tower was also tested for stiffness by applying increasing loads through a pulley, string, and weight system while measuring deflection with a dial indicator. The load was increased in 100 g increments up to 1,000 g, and the maximum measured displacement was **0.59 mm** at the largest load. Since 1,000 g is approximately 9.81 N, the experimental stiffness of the tower was calculated to be approximately **16.6 N/mm**. The final tower weight was **306.4 g**.

Overall, the project demonstrated the complete engineering design process from concept development to final testing. The final wind turbine successfully generated measurable electrical power, supported the motor at the required height, and met the major project constraints.

Through this project, our team learned how structural stiffness, blade aerodynamics, material limits, 3D printing constraints, and testing conditions all affect the performance of a wind turbine system.

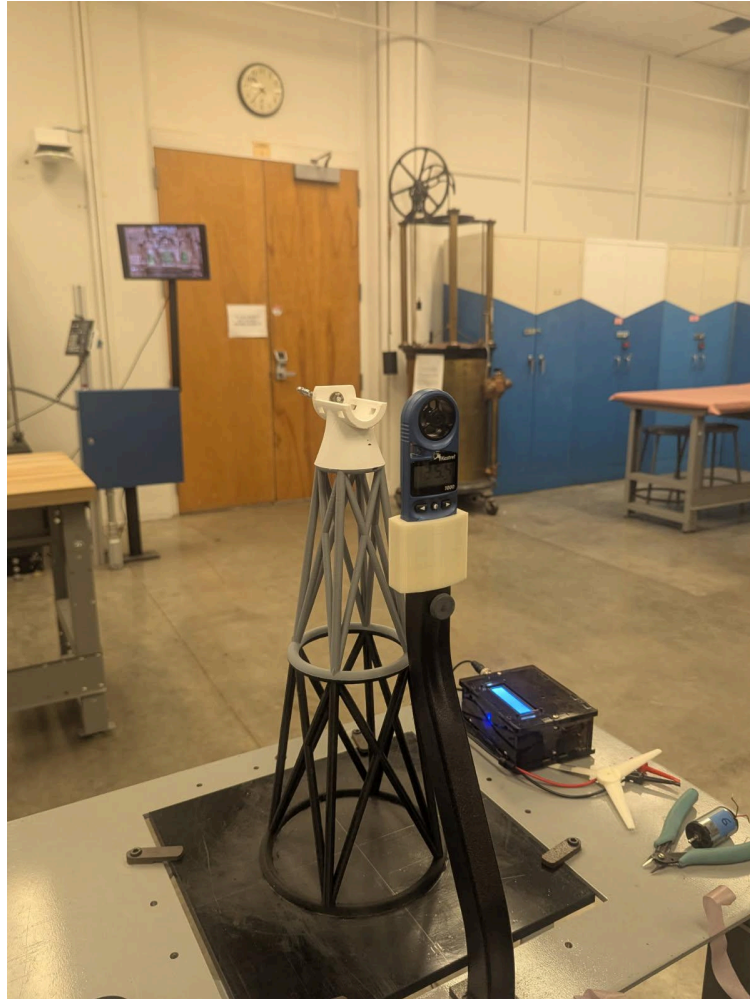


Figure 1: Final wind turbine tower assembly used for experimental stiffness testing, showing the 3D-printed tower structure, motor housing, and testing equipment.

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Introduction

Fossil fuels are the primary driver of global climate change, contributing over 68% of greenhouse gas emissions and 90% of global CO₂ emissions. Therefore, it is more important now than ever to turn towards renewable energy alternatives such as solar, hydropower, nuclear energy, and wind power. According to the US Energy Information Administration, wind power generated about 434 kWh in 2022, making up about 10.3% of US electricity generation and roughly 50-60% of renewable energy generation.

Wind power has a long history starting as early as 900 BC, utilized by Persians to propel boats, pump water and grind grain. Around 1,000 AD, wind power technology spread to European countries such as the Netherlands. These windmills resemble our modern day wind turbines, but were used primarily to drain lakes and marshes. People around the world continued to use windmills to grind grain, pump water, and cut wood, but their use in electricity generation began in the late 1800s with the rise of companies such as the establishment of the US Wind Engine Company in 1850. As demand for energy grew, wind farms were established to address energy shortages and an increasing concern for the effects of fossil fuels. Innovations in material science and engineering have led to the smooth, white wind turbines that we see today.

Wind turbines use blades to collect wind's kinetic energy. These turbines are designed with specific airfoil shapes (similar to a plane's wing) to create lift. This causes the blades to turn, turning a drive shaft connected to a generator, producing electricity. This electricity is then transferred along power cables down the length of the wind turbine and redirected to energy consumers (Figure 2).

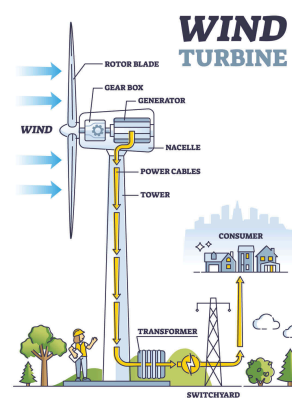


Figure 2: Wind Turbine Logistics

In this project, our team designed and fabricated a small version of a wind turbine, including a custom tower and turbine rotor blades. Our objective was to design and fabricate a wind turbine tower with the maximum stiffness when force is applied at its peak, as well as wind

turbine blades that would generate the maximum amount of energy when a 25 mph wind is applied. All non-provided parts were fabricated using the 3D printed process.

For the tower, a 12 x 12 inch lower support platform made of $\frac{3}{8}$ inch thick plastic (ABS) and a motor (which acts as a generator) were provided. The distance between the top of the platform and the center of the motor shaft must be 16 inches and the volume of material used to 3D print the tower was constrained to 17 in^3 . Our team designed both the upper platform that holds the generator (the provided motor) and the main body of the tower. Because available printers have a print space of 8 x 8 x 8 in, the body was designed in 3 parts. The three parts were then glued together, then glued to the base with a strong glue. Additionally, the motor housing was required to include a $\frac{3}{16}$ in hole for the motor shaft and no fasteners were to protrude from the bottom of the lower platform. Each member of our team sketched a tower design, then a final design was constructed using the best ideas (see **Design** for more details)

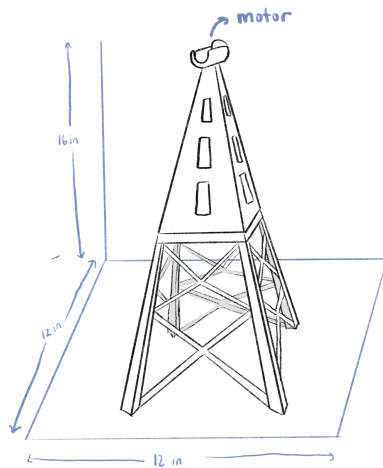


Figure 3: Jessie Design

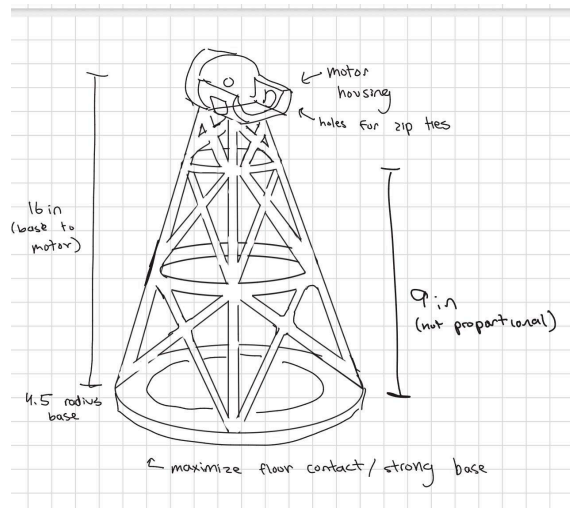


Figure 4: Kai Design

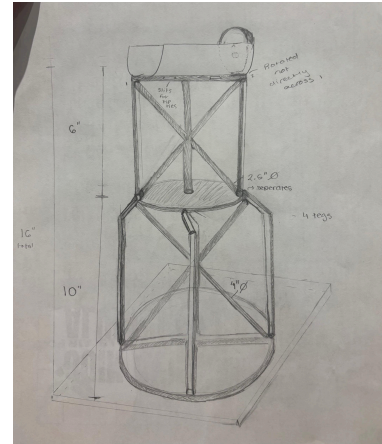
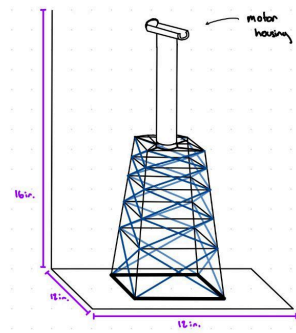
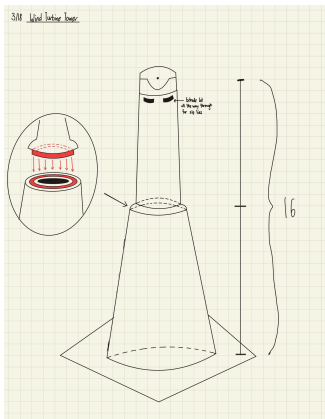


Figure 5: Brandon Design Figure 6: Sebastian Design Figure 7: Alessandra Design

For the turbine rotor blades, the diameter of the swept area must not exceed 6 inches and it must be attached to the provided motor hub CAD. Like the tower design, each member of our team sketched a rotor blade design, then a final design was constructed using the best ideas (see **Design** for more details). Each member specified a blade number, turbine shape, thickness, angle of attack, angle of twist, and other details that determine the effectiveness of a wind turbine blade.

E26 blade designs

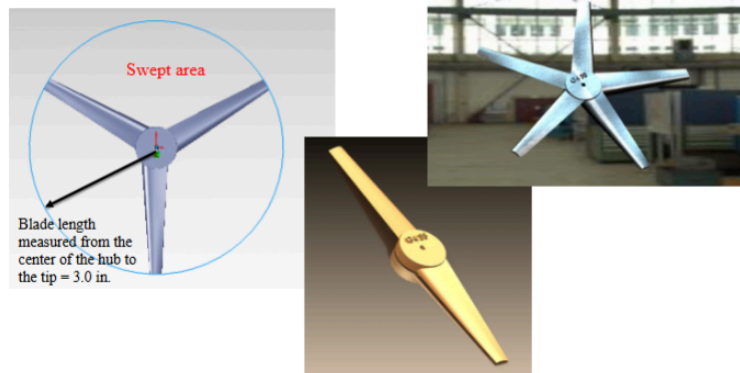


Figure 8: Blade Design Examples and Specifications

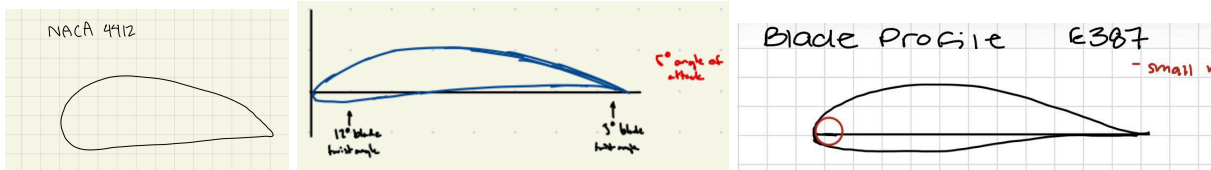


Figure 9: Various Airfoil/Blade Profile Shapes

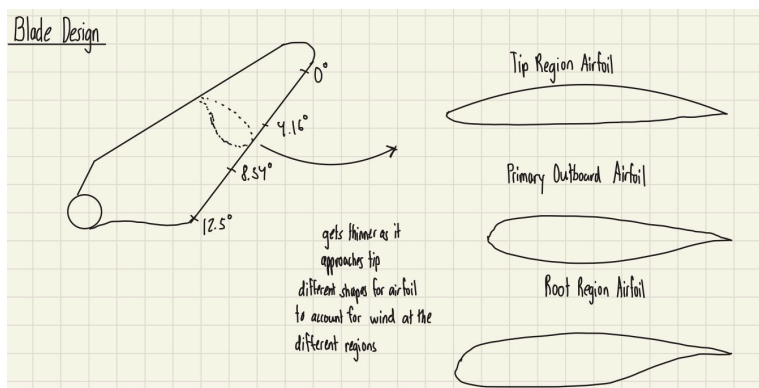


Figure 10: Blade Design 1

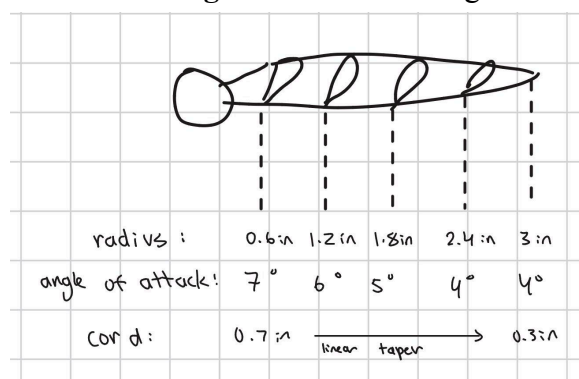


Figure 11: Blade Design 2

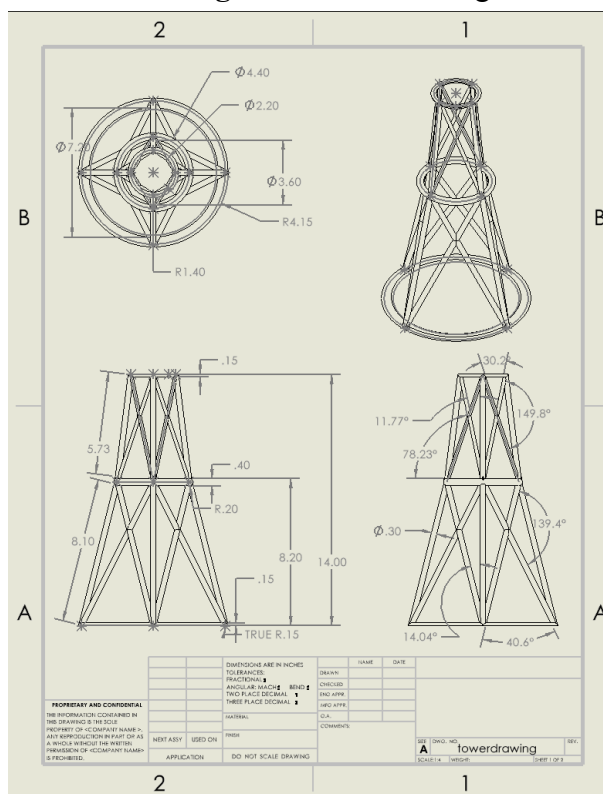


Figure 12: Final Tower Design 2D Drawings

Design

Our project uses SolidWorks as the major tool for CAD design for both the wind turbine blade and the wind turbine tower. Our project focuses on developing a wind turbine designed to be on a horizontal axis and using aerodynamic concepts that are purposefully meant to maximize performance for airfoils with a low Reynolds number. Our team proposed multiple blade designs and wind turbine structures. Ultimately, we utilized the E387 airfoil and Kai's tower structure design to maximize our wind turbines efficiency and results overall. The wind turbine blade design is composed from the E387, a cambered airfoil, specifically designated for low Reynolds number (ranging from 10,000 to 40,000). We intentionally chose a cambered airfoil over a symmetrical airfoil because it would guarantee an increase in the maximum lift coefficient. This is a crucial factor that allows our wind turbine to generate more power. The optimal angle of attack for this wind turbine was set at 5 degrees. While optimal angle of attacks range from 10-15 degrees, due to our airfoil being designated for low Reynolds number, 5 degrees is proven to create the highest ratio between the coefficient of lift to coefficient of drag. By maintaining this number, we are able to maximize aerodynamic efficiency. With this in mind, we implemented 3 blades to our design because it was able to create a balance between efficiency and stability for our small scale. Our priority in this project was efficiency over stability, and 3 blades was a well balanced design that optimized all our performance factors. To determine our blade twist angle, we took into consideration the change in relative wind along the length of the blade. By applying the Blade Element Momentum Theory (BEM) we were able to find the inflow angle, where our wind velocity meets the rotor plane. This inflow angle changes along our blade, so we used BEM Theory to determine the correct blade pitch that allows us to maintain our local angle of attack of 5 degrees. The power equation for a wind turbine blade: $P = \frac{1}{2} (\rho A v^3)$. Given that 6 in = 0.1524 m and 25 mph = 11.176 m/s our expected power is: $0.5(1.225 \text{ kg/m}^2 (\pi(0.1524 \text{ m})^2)(11.176 \text{ m/s})^3) = 62.386 \text{ W}$. Efficiency is calculated as the measured power/expected power (see Figure 26 in Results).

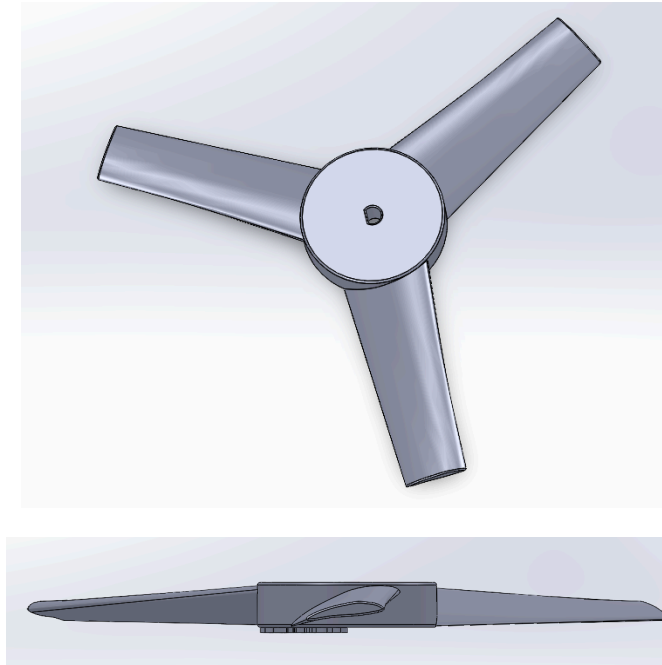


Figure 13: Final Wind Turbine Blade Design

The wind turbine tower structure we selected was influenced by multiple design features such as trusses, rings and lattice configurations. The tower design focuses on rigidity to mitigate the vibrations our wind turbine blades will generate when rotating at high speeds. Our geometric stability is determined by the base-to-height ratio and the tapered profile. The height of our structure from the 9 inch diameter base to the top of our motor housing, sits at 16 inches tall. We implemented a wide base that tapers towards the top motor housing because it will increase the stability by maximizing contact at the base and maximizing rotational inertia around the point where the force is applied. Additionally, we emphasize structural reinforcement through a series of diagonal cross braces between three vertical supports. The three main vertical supports are intended to create a clear path for load distribution.

In order to validate the structural integrity of our wind turbine tower design, we used SolidWorks to perform a Finite Element Analysis simulation. Our tower was modeled as a static system to evaluate its performance under different load conditions. The base material we selected was Medium-High Impact Acrylic, due to its ability to withstand cracking and high energy absorption.

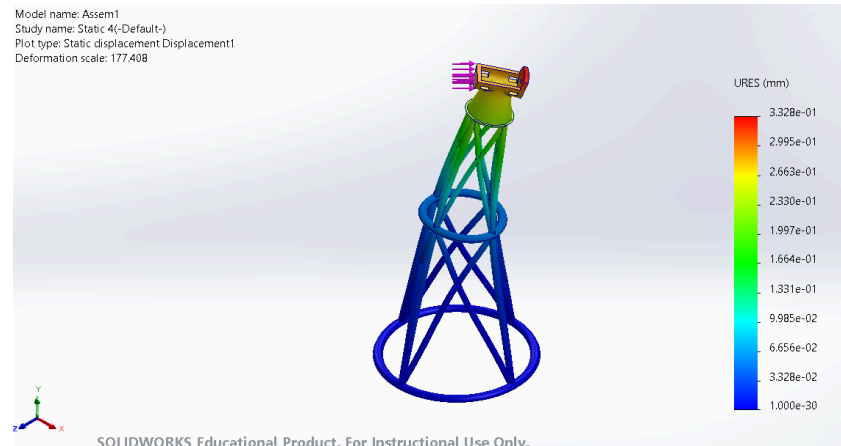


Figure 14: Static Displacement

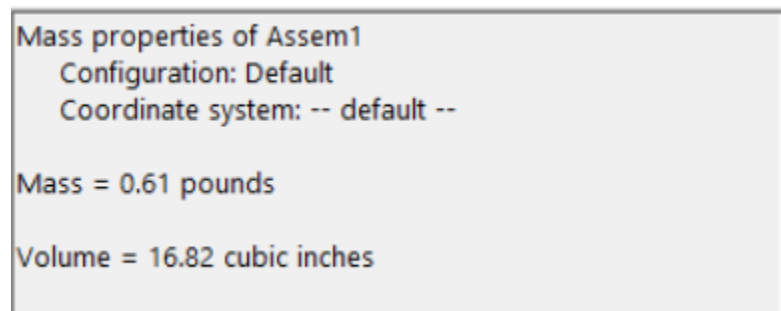


Figure 15: Mass Properties of Tower

We conducted a static analysis of the tower to evaluate how it would react with different loading conditions. Using von Mises stress criterion, we found that the maximum resultant displacement was 0.3328 mm, found at the top of the tower. Our analysis also proved structural adequacy as it maintained a minimum factor of safety of 3.0, indicating it is well within the elastic region of the material. Additionally, the stress analysis of the tower shows a maximum stress of $1.418 \times 10^6 \text{ N/m}^2$, which can be found on the top of our tower near the joints and connections. This is where the load is typically applied. The low-stress regions of our tower can be found at the bases of our tower, near the primary truss members, having stress values as low as $3.769 \times 10^{-2} \text{ N/m}^2$. Since the peak stress values of the tower are well below the materials strength, we achieve a high safety factor.

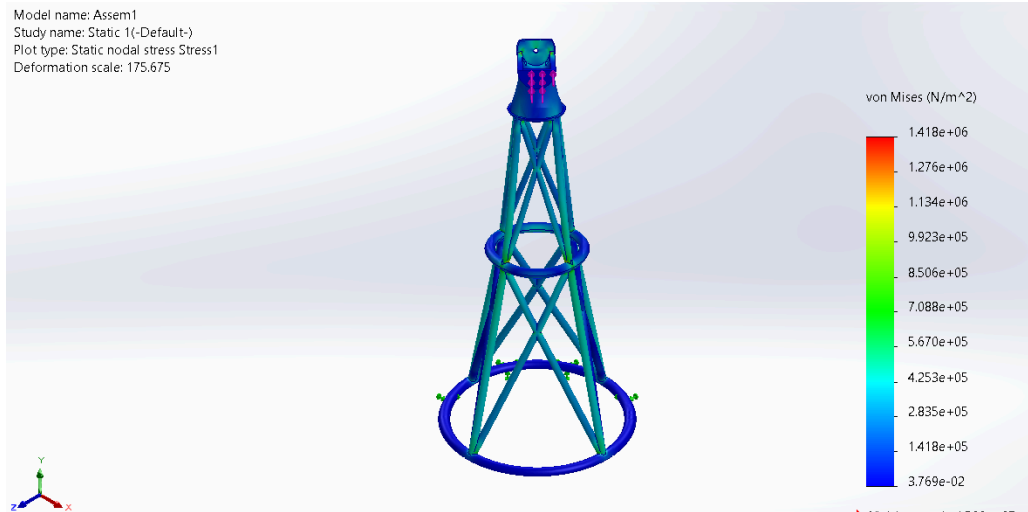


Figure 16: Stress

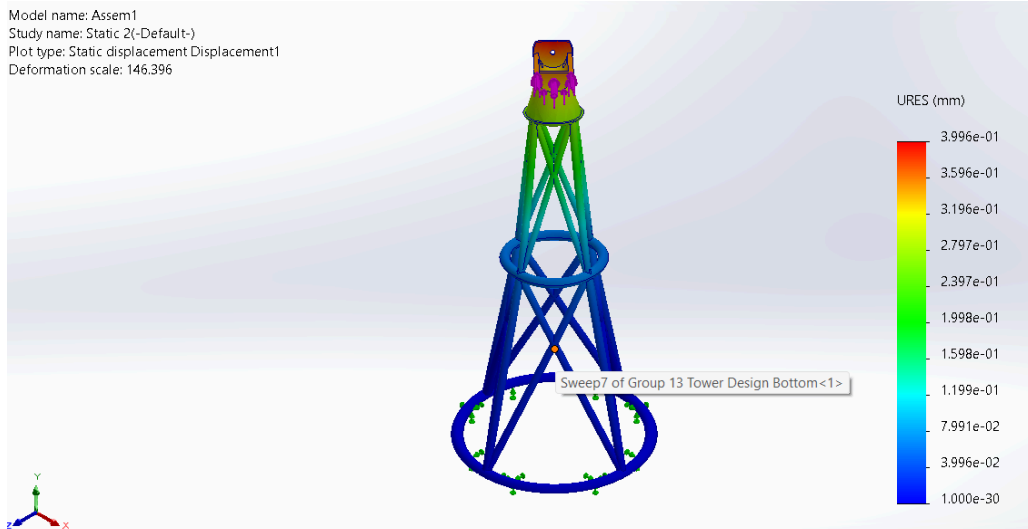


Figure 17: Displacement

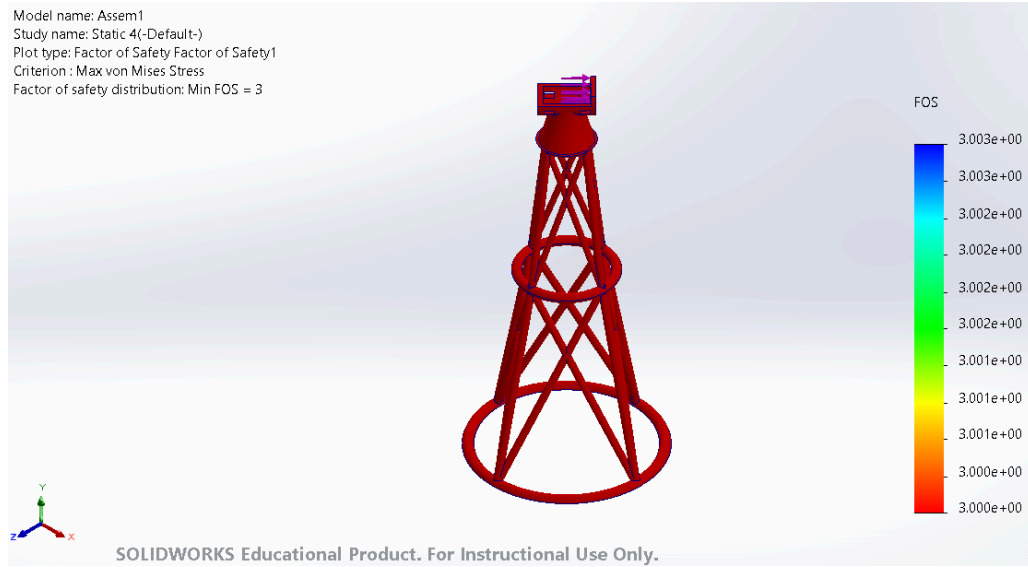


Figure 18: Factor of Safety

Load vs. Deflection

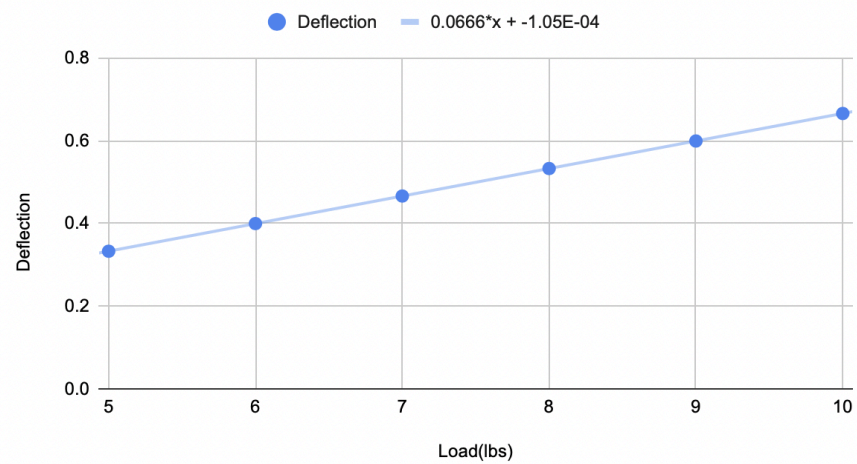


Figure 19: Load vs. Deflection

Testing and Results

As stated in the introduction, our objective in this project was to generate maximum power with our blade under 25 mph wind speeds, and test the maximum stiffness when an external force is applied to the tower. We measured the weight of our tower, which we found to be 306.4 grams. The tools used for testing power generation were an anemometer, tachometer, power meter, and blower. The blower was used to simulate wind and the anemometer was used to measure the velocity of said wind (mph). The tachometer was used to measure the rotational speed of the blade (rpm). The power meter measures the power (mW), current (mA), resistance (Ohms), and voltage (V) of the turbine.

Procedure for Power Generation Test:

1. We began the power generation testing procedure by latching the base plate of our turbine to the testing platform to hold it in place, and mounted the motor to our turbine's motor housing using zip ties. We then attached our blade to the motor, and used the power meter to determine which direction of the blade was optimal for power generation.
2. We then positioned the blower in front of our turbine so the wind velocity would be as close to 25 mph as possible, using the anemometer to measure the wind's speed.
3. After setting the wind speed, the blower was turned off, and the generator was connected to the load box using the given IC cables.
4. We then turned the blower back on and turned the potentiometer knob in small intervals, observing which combinations of current and voltage would achieve maximum power. Our team did an initial test to find the maximum power of our blade, by adjusting the potentiometer knobs on the power meter, which we found to be around 1,100 mW. Upon replicating the same procedure during actual testing, however, our maximum power fell around 1,003 mW. Maximum power in a wind turbine generator can be dependent on the loading conditions; in a real wind turbine, a computer controlled power unit constantly tweaks and adjusts the load in order to achieve maximum output. After we determined our maximum power generation, we set the power meter back to 55 Ohms and turned it slowly so that we could measure the power, current, voltage, and fan speed per 100 mW, recording them in the "Power Measurement" table.

5. At each interval, we read the current (mA), voltage (V), and power (mW) from the load box's digital display and recorded the values in our "Power Measurement" data table. Additionally, at each interval, we used the tachometer— which operates by shining a laser in the area of our blade's reflective tape to read the blade speed (rpm)— and recorded those values in the same table.
6. Lastly using our "Power Measurement" data table, we plotted graphs of our "Voltage (V) vs Current (mA)" and "Power (mWatt) vs Current (mA)" datapoints, then fit a curve to both graphs. This allowed us to determine the maximum power generated by our wind turbine.



Figure 20: Power Generation testing

Power Measurement

Data Point (#)	Voltage (V)	Current (mA)	Power (mW)	Rotational Blade Speed (rpm)
1	3.20	60.0	190.0	5509
2	3.11	97.6	249.4	5401
3	3.02	133.8	400.6	5307
4	2.57	166.5	456.8	5245
5	2.53	210.1	559.0	5119
6	2.91	213.5	657.2	5080
7	2.72	260.3	707.4	4965
8	2.62	300.6	776.2	4858
9	2.31	378.0	901.3	4650
10	2.26	439.7	1003.0	4329
11	1.06	558.4	571.4	3577
12	0.15	274.6	50.96	1378
13	0.06	271.9	12.68	947

After power generation was measured, we moved on to the next test which determined the deflection (stiffness) of our tower. The tools used for this test were an eyebolt, deflection indicator, string, and weights.

Procedure for Deflection Measurement Test:

1. We first recorded the weight of the tower, in grams, and secured the tower to the testing platform.
2. The eyebolt was already connected to the back plate of the motor housing of the tower. We attached the string to the eyebolt and pulled it over the pulley.
3. Next, we added a plastic cylinder to take the place of the motor, and secured it in place.
4. We then positioned the dial indicator and the motor mount placeholder so that they were pushed firmly together, and set the indicator to zero.
5. The weights used were 100g each and we attached them to the string one by one until a total of 1,000g (1 kg) was added to the string. At each increment, we wrote down the deflection measured by the indicator (mm) and added it to our “Deflection Measurement” table.
6. Lastly, we plotted the Load vs. Deflection, fitted a line to the points, and used the slope of the line to determine the stiffness of the wind tower.



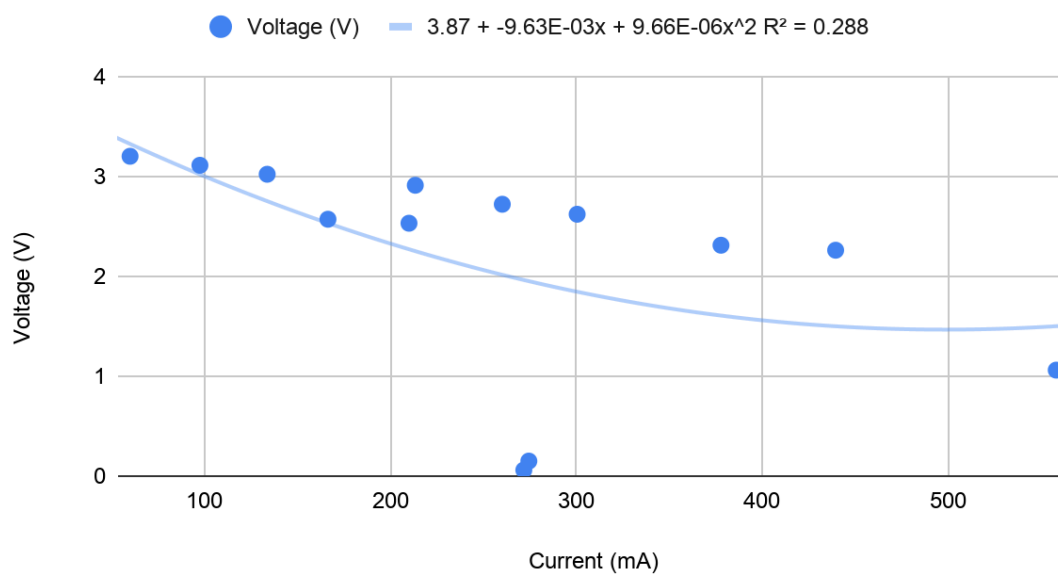
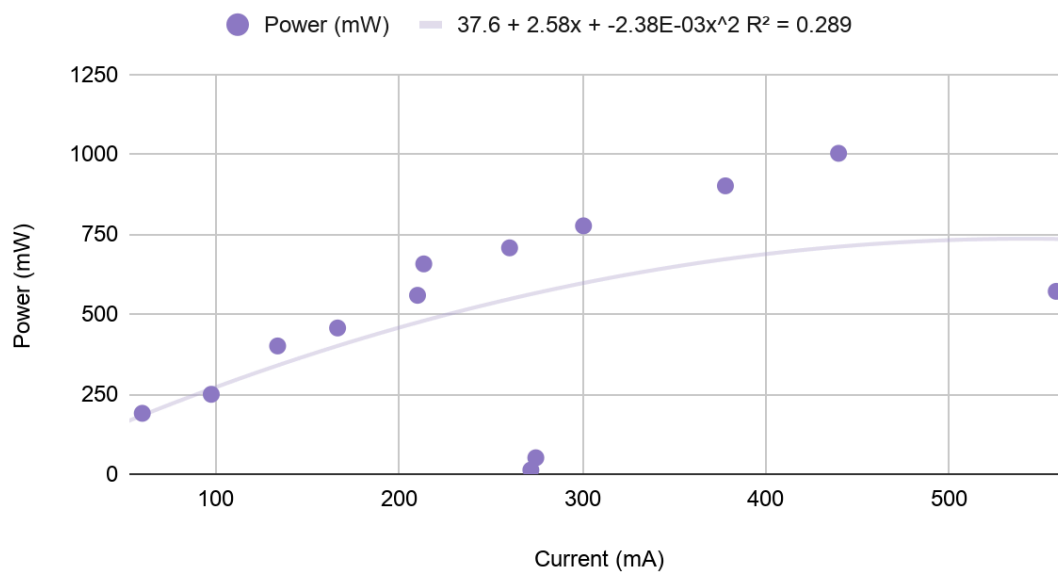
Figure 21: Deflection testing



Figure 22: Tower Failure Test

Deflection Measurement

Data Point (#)	Load (N)	Deflection (mm)	Observations
1	0.98	0.04	
2	1.96	0.10	
3	2.94	0.16	
4	3.92	0.22	
5	4.90	0.28	
6	5.88	0.34	Very linear deflection up until this point
7	6.86	0.44	Jump in deflection increase from 0.06mm to 0.10mm
8	7.84	0.48	Smaller deflection increase, but averages with the last to 0.06mm per jump when combined with the last jump
9	8.82	0.54	Return to 0.06mm deflection
10	9.80	0.59	Slightly irregular deflection increase, 0.06mm to 0.05mm

Plots:**Voltage (V) vs. Current (mA)****Figure 23: Voltage (V) vs. Current (mA)****Power (mW) vs. Current (mA)****Figure 24: Power (mW) vs. Current (mA)**

Load (N) vs. Deflection (mm)

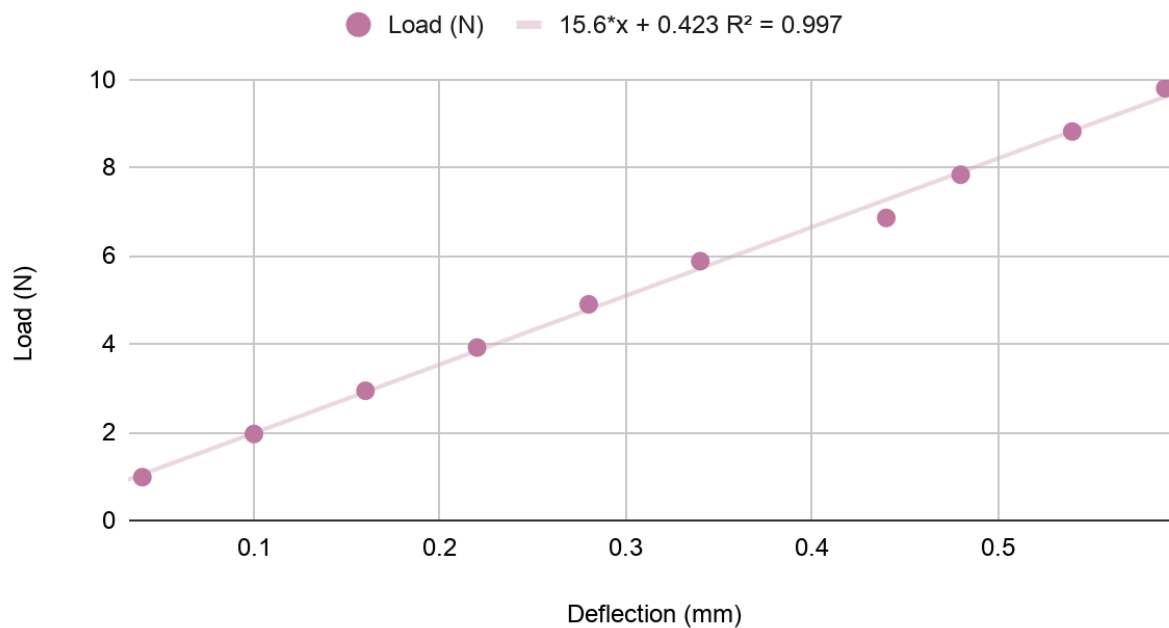


Figure 25: Load (N) vs. Deflection (mm)

Tower Stiffness: 15.6

Efficiency vs. Current (mA)

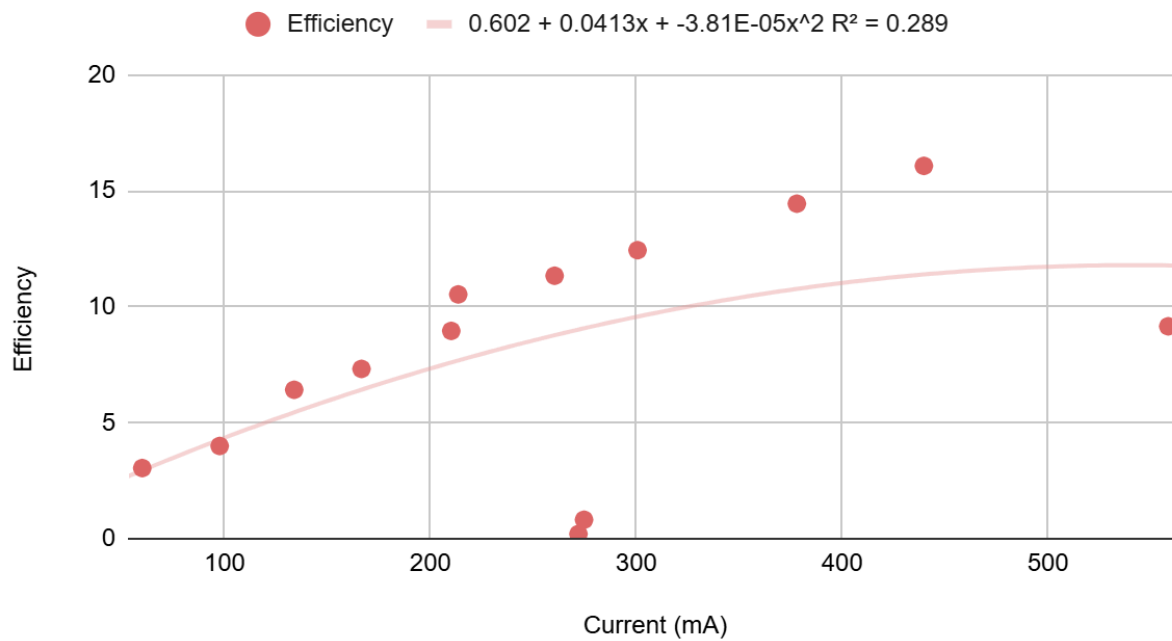


Figure 26: Efficiency vs. Current (mA)

CAD Drawings

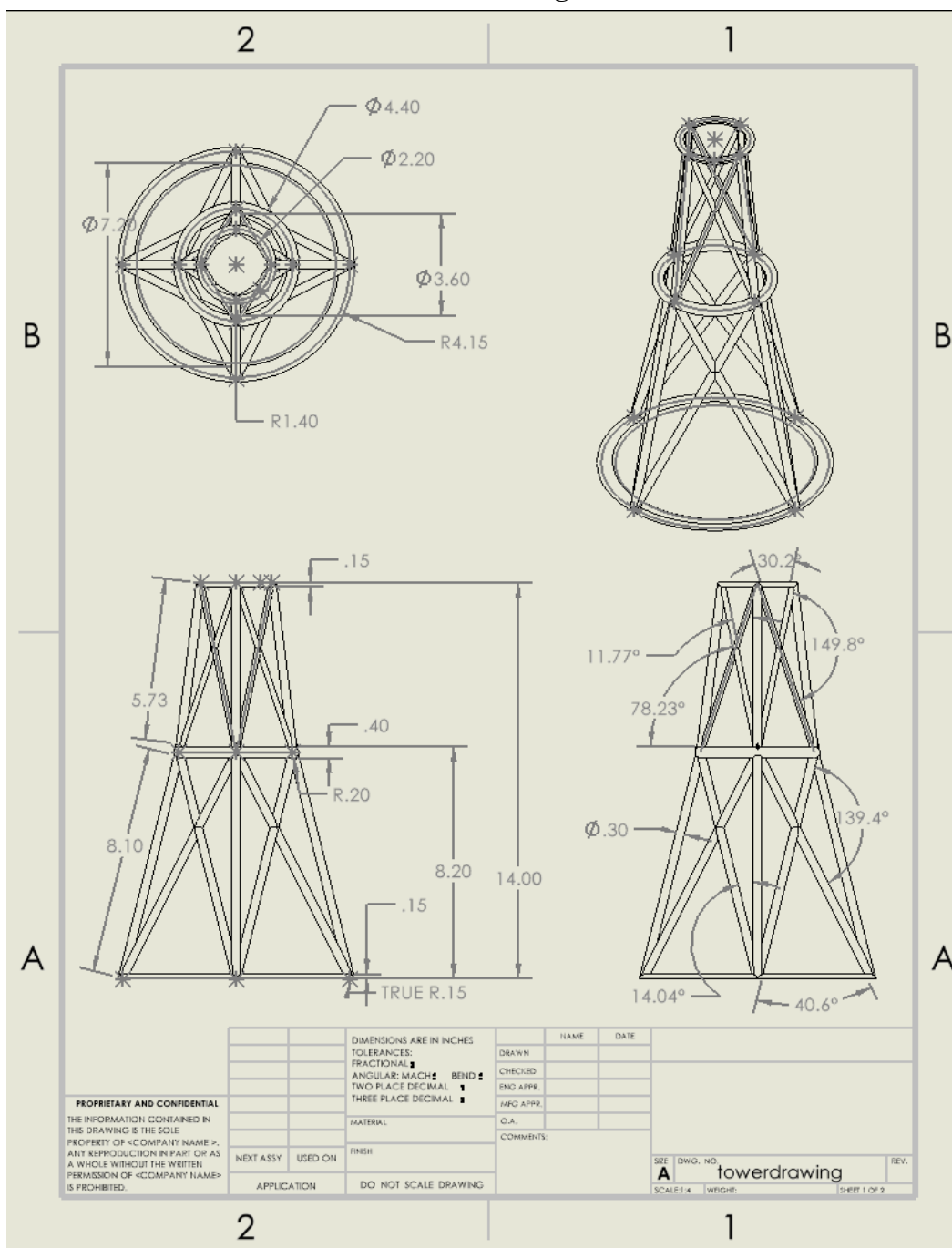


Figure 27: 2D CAD Drawing of Tower Base

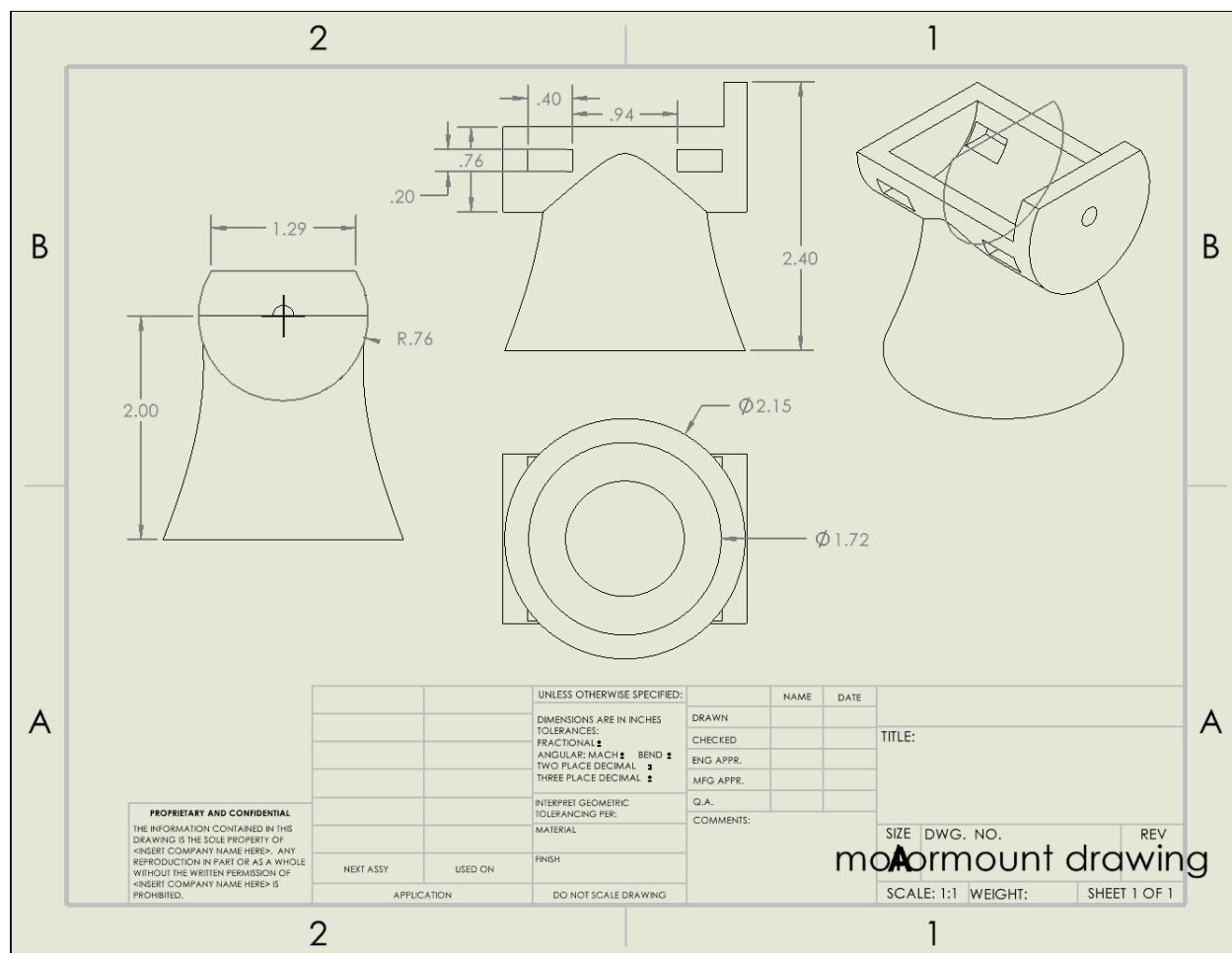


Figure 28: 2D CAD Drawing of Motor Mount

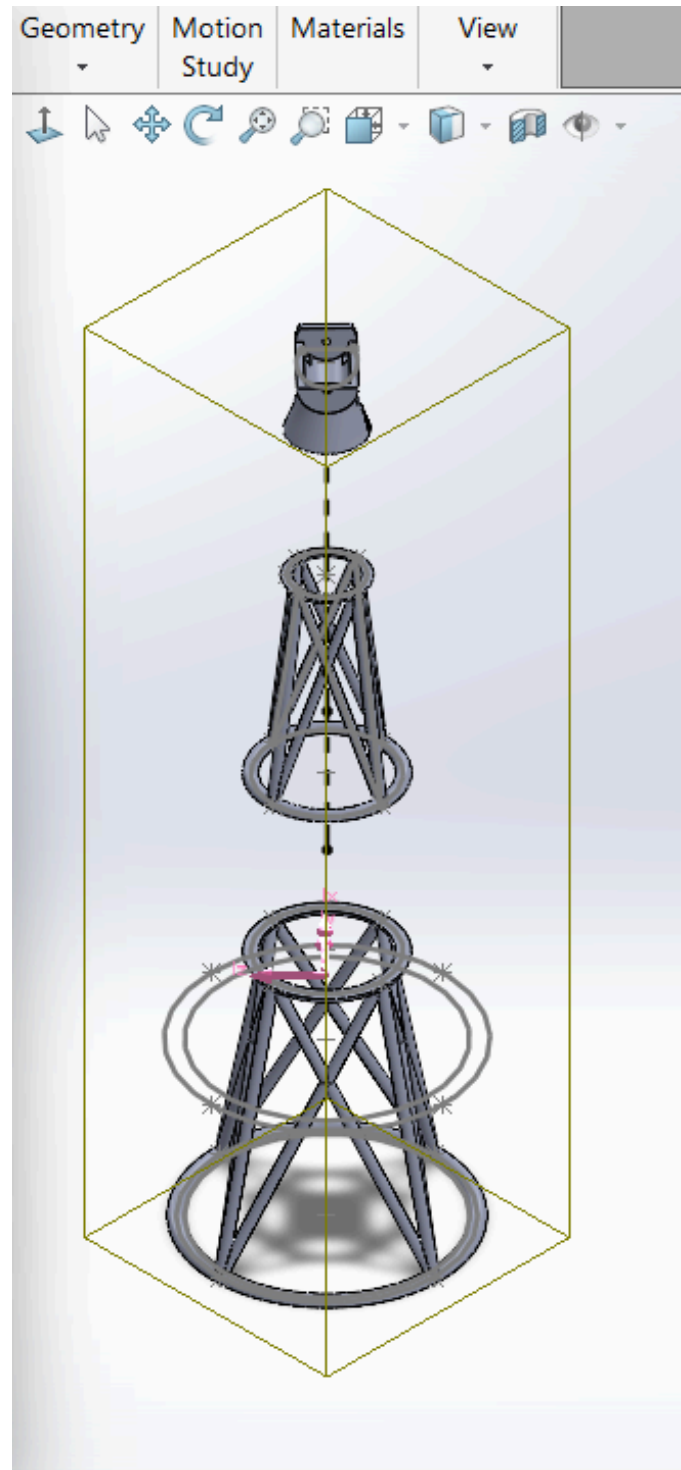


Figure 29: 3D Assembly of Tower

Conclusions

In this project, our team designed, fabricated, and tested a small-scale wind turbine system consisting of a custom 3D-printed tower and rotor blade assembly. The main goals were to maximize the stiffness of the tower under loading and maximize the power generated by the blades under a 25 mph wind condition. Using initial design sketches, SolidWorks CAD modeling, 3D printing, and experimental testing, we developed a final design that satisfied the major project constraints, including the required 16-inch motor shaft height, 6-inch blade swept diameter, and material volume limit.

The final turbine performed successfully during testing. The blade assembly generated a maximum power output of 1,003 mW, showing that the design was able to convert wind energy into electrical energy through the provided motor generator. The tower had a total weight of 306.4 g and reached a maximum deflection of 0.59 mm under a 1,000 g load. Based on this load and displacement, the experimental tower stiffness was calculated to be approximately 16.6 N/mm.

This project helped us better understand the connection between design decisions and real-world performance. For the blade design, we learned that airfoil shape, blade twist, angle of attack, and blade geometry directly affect rotational speed and power generation. For the tower design, we learned that structural geometry, cross bracing, material placement, and unsupported vertical lengths have a major effect on stiffness and deflection. We also learned that CAD models and simulations are useful design tools, but final performance can still be affected by 3D printing quality, assembly methods, glue joints, and testing conditions.

Overall, the project was successful because our wind turbine met the required design constraints, generated a strong power output, and allowed us to evaluate the stiffness of our tower through experimental testing. The results also showed areas for future improvement, especially in optimizing the blade airfoil for low-speed wind conditions and adding more effective cross bracing to further reduce tower deflection. These findings would help guide future design changes and improve both the power generation and structural performance of the turbine.

Recommendations for Future Work

Although the results for our wind turbine were good, there are definitely improvements that could be made. For the airfoil of the blade, we used the E387 blade model as a reference. While it performed well, we should have modeled our blade after the SG6043 blade models because they were engineered to have exceptional lift-to-drag ratios at low Reynolds numbers which would consequentially improve the rotational speed and power generation. According to Emmanuel Yeboah Osei, the highest lift-to-drag ratio at lower Re for the SG6043 airfoil. Changing the twist angle or angle of attack aren't pressing matters because they don't change for different blade models with a diameter of 6 inches. To improve the deflection of our tower, we should have added more cross bracing. Adding more cross bracing would get rid of long unsupported verticals, which are detrimental for stiffness.

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